Theoretical Module : Solar Flare Mechanics

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Task 1 – The Sun's Explosive Energy: Solar Flares

(a) What does it mean for magnetic field lines to "store energy"? Explain with an analogy.

From what I understood, when scientists say that magnetic field lines "store energy", they do not mean that the field lines themselves are physical objects holding energy inside them. It is actually the magnetic field around the Sun that becomes twisted and stressed because of the continuous movement of hot plasma on the Sun's surface. The Sun is made of extremely hot ionized gas, and this plasma is always moving because of convection. As the plasma moves, it drags the magnetic field lines along with it. As the magnetic field becomes more twisted, more magnetic energy gets stored inside it. This stored energy is what later gets released suddenly during magnetic reconnection, producing a solar flare. Over time, these field lines become more and more twisted, stretched and tangled. This process stores magnetic energy in that region.

The easiest way I can understand this is by thinking about a stretched spring or a rubber band. When we stretch a spring, we are putting energy into it. As long as we keep holding the spring, the energy remains stored inside because of the tension. The more we stretch it, the more energy gets stored. But if we suddenly let go, the spring quickly returns to its original shape and all that stored energy is released almost instantly.

I think the same thing happens on the Sun. The magnetic field lines keep getting twisted because the plasma underneath is always moving. They cannot keep storing energy forever. At some point they become unstable and suddenly reconnect into a simpler and more stable arrangement. During this process, the magnetic energy that had been stored for a long time gets released within a very short time. This sudden release of energy is what we observe as a solar flare. So instead of slowly releasing energy, the Sun releases a huge amount of energy almost all at once.

(b) Describe the three effects of a solar flare.

When a solar flare occurs, the stored magnetic energy is released very rapidly. From what I learned, this released energy does not go into just one process. Instead, it gets divided into different forms, and each one affects the Sun and the surrounding space in a different way.

The first effect is that the surrounding plasma becomes extremely hot. Before reading about

solar flares, I knew the Sun was already very hot, but I did not realize that during a flare some regions can become several times hotter than normal. The released energy heats the nearby gas to temperatures of several million degrees Celsius. Because of this enormous temperature, the region becomes much brighter and emits large amounts of X-rays and ultraviolet radiation. This is one of the reasons why solar flares can easily be detected by space telescopes.

The second effect is the acceleration of charged particles such as electrons and protons. The energy released during magnetic reconnection acts almost like a giant natural particle accelerator. Instead of simply heating the gas, a part of the energy pushes these charged particles to extremely high speeds, sometimes close to the speed of light. These energetic particles can escape from the Sun and travel through interplanetary space. If Earth lies in their path, they can interact with our planet's magnetic field and atmosphere. This is one of the main reasons why space weather becomes important for satellites and astronauts.

The third effect is that huge clouds of plasma can be thrown away from the Sun into space. These clouds contain billions of tons of hot gas along with their own magnetic fields. When this material moves away from the Sun, it is known as a Coronal Mass Ejection (CME). Although solar flares and CMEs are not exactly the same thing, they often occur together during strong solar activity. If a CME is directed towards Earth, it can disturb Earth's magnetosphere and produce geomagnetic storms. These storms are responsible for beautiful auroras near the poles, but they can also create problems for satellites, communication systems and electrical infrastructure.

So, a solar flare does much more than simply produce light. It heats the surrounding plasma, accelerates charged particles to very high speeds and can also eject enormous amounts of solar material into space. All these effects together make solar flares one of the most energetic events in our Solar System.

(c) Why can solar flares disrupt satellites, GPS and power grids?

From what I understood, modern technology depends heavily on satellites, radio communication and electrical systems. When a strong solar flare occurs, it releases large amounts of high-energy radiation and energetic charged particles. These particles travel through space and eventually interact with Earth's magnetic field. This interaction disturbs the magnetosphere and the upper atmosphere, creating what scientists call space weather.

I think one major reason electronic systems get affected is because these magnetic disturbances can induce unwanted electric currents in long conductors such as power transmission lines. If these currents become strong enough, they can overload transformers and damage parts of the electrical grid. Satellites are also directly exposed to energetic particles because they are outside most of Earth's atmosphere. These particles can interfere with onboard electronics, affect communication signals and sometimes even damage sensitive instruments. GPS signals also pass through the Earth's ionosphere, and during periods of intense solar activity this region becomes highly disturbed. Because of that, GPS signals may

get delayed or distorted, reducing their accuracy.

This is why scientists continuously monitor solar activity. Even a few hours of advance warning can help satellite operators, power companies and communication networks prepare for severe space weather events and reduce possible damage.

Task 2 – Magnetic Fields and the Stability of Solar Structures

(a) Think about winding up a rubber band around your fingers. Describe, in everyday language, what happens physically as you twist it more and more, and why it eventually snaps or flies off. How is this a useful analogy for a twisted magnetic loop on the Sun?

From what I understood, a rubber band is a very simple example that helps explain how energy can be stored in a system. When I first start twisting a rubber band around my fingers, nothing unusual happens. It simply becomes tighter and stores a small amount of energy. If I continue twisting it, the rubber band becomes more and more stressed. I can actually feel the tension increasing because it naturally wants to return to its original shape. The more twists I add, the more unstable it becomes.

After some point, the rubber band cannot stay in that twisted shape anymore. It starts forming loops, bends into strange shapes, or sometimes suddenly slips off my fingers. If it is stretched too much, it may even snap. When that happens, all the energy that was stored because of the twisting gets released very quickly. The whole process happens almost instantly because the rubber band is trying to reach a more stable and relaxed state.

I think this is a very good analogy for the magnetic loops present in the Sun's corona. The magnetic loops are anchored to the Sun's surface, but the hot plasma below the surface is always moving due to convection. These continuous motions slowly twist the magnetic field lines, just like my fingers twist the rubber band. As the twisting continues, more magnetic energy gets stored inside the loop. For some time the loop remains stable, but if the twisting becomes too large, it can no longer maintain its original shape. The loop begins to buckle and forms a helical or spiral structure. This is known as the kink instability. Finally, the stored magnetic energy is released suddenly, producing a powerful solar eruption or flare.

So, in both cases, energy is stored gradually while twisting increases, but once the system becomes unstable, it suddenly releases that energy by moving into a more stable configuration.

(b) Imagine holding a beach ball underwater. As long as the pressure of the water is strong enough, the ball stays submerged. If you move the ball to shallower water (where the pressure is lower), it becomes harder to hold down. Explain how this is a useful analogy for the torus instability of a solar magnetic loop. In your answer, identify what the "beach ball", the "water pressure", and the "shallower water" each represent in the solar context.

I found the beach ball analogy much easier to understand because it explains why a magnetic loop suddenly erupts even though it was stable before.

If I hold a beach ball deep underwater, I have to push it down because it naturally wants to float upward. At greater depth, the surrounding water pressure helps keep the ball underwater, so it is easier to control. But if I slowly move the beach ball towards the surface, the surrounding pressure becomes weaker. Now the ball wants to float upward even more strongly, and it becomes harder to keep it underwater. Eventually, I cannot hold it anymore, and it suddenly shoots upward and comes out of the water.

In this analogy:

The beach ball represents the twisted magnetic loop present in the Sun's corona.

The water pressure represents the overlying magnetic field, which keeps the magnetic loop confined near the Sun's surface.

The shallower water represents higher regions of the corona, where the overlying magnetic field becomes weaker.

As long as the overlying magnetic field is strong enough, it can hold the magnetic loop in place even though the loop is trying to expand outward. But as the loop rises to greater heights, the surrounding magnetic field becomes weaker. At some point, it is no longer strong enough to confine the loop. Just like the beach ball suddenly escapes from the water, the magnetic loop also escapes rapidly into space. This sudden upward eruption is called the torus instability.

I think this analogy is very useful because it clearly shows that the loop is not exploding randomly. Instead, it is the weakening of the magnetic field above it that allows the loop to break free and erupt.

(c) Why might it matter whether a magnetic loop erupts or stays confined? Give one example of a consequence that an eruption could have outside the Sun.

From what I understood, whether a magnetic loop erupts or remains confined is very important because it determines how much of the released energy actually leaves the Sun and affects the surrounding space.

If the magnetic loop remains confined, most of the energy stays close to the Sun. A solar flare

may still occur, but very little solar material escapes into interplanetary space. In this case, the effects on Earth are usually much smaller because the eruption remains mostly localized.

However, if the magnetic loop erupts successfully, it can throw a huge amount of hot plasma and magnetic field into space as a Coronal Mass Ejection (CME). If this CME is travelling towards Earth, it can interact with Earth's magnetic field and produce a geomagnetic storm.

One important consequence of such an eruption is that it can disturb satellites and GPS systems. Satellites contain sensitive electronic instruments, and energetic charged particles from the Sun can interfere with their operation or even damage some components. GPS signals also travel through Earth's upper atmosphere, which becomes disturbed during strong space weather events. As a result, navigation systems may become less accurate and communication systems may also experience interruptions.

Because of these possible effects, scientists carefully monitor the Sun's magnetic loops and try to predict whether they are likely to remain stable or erupt. Even a short warning before a major eruption can help satellite operators and other organisations prepare for the incoming space weather.

Task 3 – Why Surface Measurements Tell Us About the Corona

(a) Think about the skipping rope analogy. If one person holding the rope starts moving their hands in circles, what happens to the rope between them? How does this describe the way that motions at the solar surface can inject energy into the overlying corona?

From what I understood, the skipping rope analogy is probably one of the easiest ways to imagine what happens on the Sun. Suppose two people are holding the ends of a skipping rope. If both of them keep their hands still, the rope also stays almost still. But if one person suddenly starts moving their hands in circles or twisting the rope, the movement immediately travels through the rope. The rope starts twisting, bending and storing energy because one end is constantly being disturbed. If this continues for a long time, the rope becomes more and more tangled.

I think the magnetic field lines on the Sun behave in a very similar way. The ends of these magnetic loops are attached to the Sun's surface, which is called the photosphere. The plasma below the photosphere is always moving because of convection. These continuous motions drag, twist and stretch the magnetic field lines. Since the field lines are anchored at the surface, all this motion gets transferred into the corona above.

Even though the corona is located much higher than the photosphere, the energy is actually supplied from below. As the surface keeps moving, the magnetic loops store more and more

energy. Over time the loops become highly stressed and unstable. Eventually the magnetic field reconnects into a simpler arrangement, releasing the stored magnetic energy as a solar flare. So, just like the skipping rope carries motion from one person's hands to the rest of the rope, the magnetic field carries energy from the Sun's surface into the corona.

(b) In physics, pressure tells us which "side" dominates an interaction. Explain what β << 1 implies about whether the gas or the magnetic field is in control. Then explain what β >> 1 implies. Where on the Sun is each condition found?

From what I understood, plasma beta (β) is the ratio of gas pressure (P gas) to magnetic pressure (P magnetic). It tells us whether the gas pressure or the magnetic pressure is more dominant in a particular region of the Sun.

When β << 1, the magnetic pressure is much larger than the gas pressure. In this situation, the magnetic field is in complete control. The hot plasma cannot move freely in any direction and instead follows the magnetic field lines. Because of this, the magnetic loops remain well organised and most of the activity is controlled by magnetic forces. This condition is mainly found in the solar corona, where the gas density is low but the magnetic field is still strong.

On the other hand, when β >> 1, the gas pressure becomes much stronger than the magnetic pressure. Here the plasma is dominant. The moving gas can push, drag and twist the magnetic field lines as it moves around. Instead of the magnetic field controlling the plasma, the plasma now changes the shape of the magnetic field. This condition is mainly found in the photosphere, where the gas is much denser and convection is continuously taking place.

I think this difference between β << 1 and β >> 1 explains why the photosphere is responsible for twisting the magnetic field, while the corona mainly follows whatever shape the magnetic field takes.

(c) Briefly explain why measurements of the photospheric magnetic field—even though the flare happens much higher up—still give scientists useful information about where and when a flare might occur.

At first this confused me because scientists mainly measure the magnetic field at the photosphere, while solar flares actually happen much higher in the corona. After reading about it, I understood that the magnetic loops connecting these two regions are the main reason.

The magnetic loops responsible for solar flares are anchored to the photosphere through their footpoints. Whenever the plasma at the surface moves, it drags these footpoints along with it. This gradually twists, stretches and stresses the magnetic loops extending into the corona. So even though the flare itself occurs high above the surface, the conditions that create the flare begin at the photosphere.

By measuring the magnetic field at the surface, scientists can identify regions where the magnetic field is becoming highly twisted or where opposite magnetic polarities are packed

close together. These are signs that a large amount of magnetic energy is being stored. Such active regions have a much greater chance of producing solar flares in the future.

So, I think the photosphere works like an early warning system. The flare may happen in the corona, but the changes happening at the surface give scientists important clues about where magnetic energy is building up and which active regions are most likely to become unstable.

Task 4 – Reading the Sun's Magnetic "Warning Signs"

(a) Using only the descriptions above (no equations required), explain in plain language what the magnetic field of Active Region A probably looks like compared to Active Region B. Which region would you expect to be more likely to produce a large solar flare, and why?

According to the table, Active Region A has much larger values of R_VALUE, TOTPOT, TOTUSJH and TOTUSJZ than Active Region B. This immediately tells me that the magnetic field in Region A is much more active, stressed and complicated than the magnetic field in Region B.

The first parameter is R_VALUE, which tells us how closely opposite magnetic polarities are packed together. Since Active Region A has a very large R_VALUE, I think positive and negative magnetic fields are extremely close to each other. This makes it much easier for magnetic reconnection to happen, which is the main process behind solar flares.

The second parameter is TOTPOT, which measures how much free magnetic energy is stored in the region. A high value means that the magnetic field is carrying a large amount of extra energy that has not yet been released. I imagine it like a stretched spring that has been pulled very tightly. The more energy it stores, the more powerful the release can be once it becomes unstable.

The third parameter is TOTUSJH, which represents how twisted or helical the magnetic field lines are. A large value tells me that the magnetic loops are no longer simple or smooth. Instead, they have become highly twisted because of the continuous movement of plasma on the Sun's surface. Twisted magnetic fields usually store much more energy than straight field lines.

The fourth parameter is TOTUSJZ, which measures the total vertical electric current in the active region. Since electric currents and twisted magnetic fields are closely related, a high value of TOTUSJZ suggests that strong electric currents are flowing through the magnetic loops. This also indicates that the region is highly stressed.

In comparison, Active Region B has very small values for all these parameters. This means the

opposite magnetic polarities are not packed closely together, very little free magnetic energy is stored, the magnetic field lines are much less twisted and only weak electric currents are present. Overall, the magnetic field is much simpler and more stable.

Because of all these reasons, I would definitely expect Active Region A to be much more likely to produce a large solar flare than Active Region B. It stores more energy, has stronger electric currents, more twisted magnetic fields and a much higher chance of magnetic reconnection.

(b) Consider an analogy: a calm, flat lake versus a lake with large waves, strong currents, and a whirlpool at the centre. Which lake stores more mechanical energy? How does this analogy map onto the difference between Active Region B and Active Region A?

I think the lake with large waves, strong currents and a whirlpool stores much more mechanical energy than the calm lake. A calm lake has very little motion, so there is very little energy available. On the other hand, the rough lake is constantly moving, and all those waves and currents show that a large amount of energy is already stored in the water.

I can relate this analogy to the two active regions on the Sun quite easily. The calm lake is similar to Active Region B. Since all four magnetic parameters are small, the magnetic field is relatively simple and stable. The field lines are not highly twisted, electric currents are weak and only a small amount of magnetic energy is stored. Because of this, the region is unlikely to produce a strong solar flare.

The rough lake with waves and a whirlpool is like Active Region A. A large R_VALUE means opposite magnetic fields are packed closely together. A large TOTPOT means a huge amount of magnetic energy has already been stored. High TOTUSJH tells us that the field lines are highly twisted, while a large TOTUSJZ indicates strong electric currents flowing through the region. All these conditions together make the region much more unstable. Just like the rough lake contains much more energy than the calm lake, Active Region A contains much more magnetic energy than Active Region B and is therefore much more likely to produce a powerful solar flare.

(c) These parameters are measured at the solar surface, but the flare happens in the corona above it. Explain in one or two sentences why a surface measurement can still act as a reliable warning sign for activity in the corona.

From what I understood, the magnetic loops responsible for solar flares are firmly anchored to the photosphere through their footpoints. So, even though the flare occurs in the corona, measurements of the surface magnetic field reveal how much twisting, stress and magnetic energy are building up in those loops, making them reliable warning signs for future solar activity.

(d) Machine-learning models that use these parameters achieve better-than-chance forecasting of large solar flares, but they are not perfectly accurate. Suggest one physical reason why predicting exactly when a flare will occur is genuinely difficult, even with excellent measurements.

I think one major reason is that the Sun is an extremely dynamic and constantly changing system. Even if scientists measure the magnetic field very accurately, they can only estimate how much magnetic energy has been stored. They still cannot know the exact moment when the magnetic field will become unstable enough for magnetic reconnection to occur.

I imagine it like twisting a rubber band. If I keep twisting it, I know that eventually it will snap or suddenly slip away, but it is almost impossible to predict the exact second when that will happen. The same thing happens on the Sun. Scientists may know that an active region has all the conditions needed for a large solar flare, but many small plasma motions and magnetic interactions are happening continuously. These tiny changes are difficult to measure completely, making the exact timing of a flare very hard to predict. That is why machine-learning models improve the chances of forecasting solar flares, but they still cannot predict every event with perfect accuracy.

Computational Module : The M5 Flare Problem

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Report: M5+ Solar Flare Prediction Using Machine Learning

1. Key Findings from Data Exploration

Before building any machine learning model, I first explored the training dataset to understand its structure and quality. This helped me identify any problems in the data before moving to model training.After removing the rows labelled as padding, the cleaned dataset contained 135,034 observations with 13 columns. I checked for missing values and duplicate records but found none, so only a small amount of preprocessing was required.

The Flare column contained three values: P, N, and padding. Since padding rows were not actual observations, I removed them. The remaining values were converted into a binary target variable where P represented an M5+ flare (1) and N represented no flare (0).

One of the biggest challenges I noticed was the severe class imbalance. There were around 123,840 non-flare observations and only about 11,194 flare observations, giving an imbalance ratio of nearly 11:1. This showed that accuracy alone would not be enough to evaluate the models because a model predicting only the majority class could still achieve high accuracy. For this reason, I focused more on Recall, Precision, F1-score, and ROC-AUC during model evaluation.

Next, I explored the numerical features. Since the magnetic field measurements were already normalized between -1 and 1, additional scaling was unnecessary. I also looked at the distribution of each feature and found that most of them were skewed rather than normally distributed, which is expected because strong solar activity is relatively rare.

To understand which features were useful, I performed a correlation analysis. Parameters such as TOTUSJH, TOTPOT, ABSNJZH, TOTUSJZ, USFLUX, and R_VALUE showed moderate positive correlation with flare occurrence, suggesting that they carry useful information for prediction. In contrast, MEANALP showed very little correlation with the target variable.I also observed that NOAA_AR and HARPNUM are identification numbers for solar active regions rather than physical measurements. Since they do not describe the magnetic properties of the Sun, I did not treat them as important predictive features.

Overall, the exploratory analysis showed that the magnetic field parameters contain useful

information for predicting M5+ flares. At the same time, it highlighted that handling the class imbalance would be one of the most important parts of the project.

2. Best Performing Model and Why I Selected It

After completing the exploratory analysis, I moved on to feature engineering, handling class imbalance, and training different machine learning models.I first trained a baseline Decision Tree model using the original dataset. Although it produced a good accuracy, its ability to detect actual flare events was poor because of the highly imbalanced data. This confirmed that relying only on accuracy would give a misleading picture of model performance.

To improve the prediction of the minority class, I applied SMOTE (Synthetic Minority Over-sampling Technique). SMOTE generates synthetic samples for the minority class instead of simply duplicating existing ones, allowing the model to learn a better decision boundary. To avoid data leakage, SMOTE was applied only to the training data using an imbalanced learning pipeline.

I also created five new features from the existing magnetic field measurements. These features were based on physical ideas such as magnetic flux density, normalized helicity, vertical current density, along with two time-related features extracted from the observation timestamp. These engineered features helped provide additional information that was not directly available in the original dataset.

I then trained and compared five tree-based machine learning algorithms: 1)Decision Tree 2)Random Forest 3)AdaBoost 4)Gradient Boosting 5)XGBoost

Each model was tuned using GridSearchCV together with Stratified K-Fold Cross Validation. Since the main objective of this project was to detect as many M5+ flares as possible, Recall was used as the scoring metric during hyperparameter tuning.The final performance of each model on the testing dataset is given below.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Decision Tree	0.8994	0.0363	0.2165	0.0622	0.7576
Random Forest	0.9568	0.0293	0.0563	0.0386	0.7505
AdaBoost	0.8987	0.0099	0.0563	0.0168	0.6617
Gradient Boosting	0.9433	0.0507	0.1515	0.0760	0.7498
XGBoost	0.9222	0.0402	0.1775	0.0656	0.7427

After comparing all five models, I selected the Decision Tree as the final model. Initially, I expected models like Random Forest or XGBoost to perform better, but the Decision Tree

achieved the highest Recall among all models. Since the project clearly states that missing a flare is more costly than producing a false alarm, Recall was the most important metric. Even though the Precision was relatively low, the Decision Tree detected more actual flare events than the other models, making it the best choice for this problem.

3. Number of Flares Correctly Detected

The testing dataset contained 231 actual M5+ solar flares.The final Decision Tree model achieved a Recall of 0.2165, which means it correctly detected approximately 21.65% of all flare events. The number of correctly detected flares is: 231 × 0.2165 ≈ 50 flares

Therefore, the model successfully detected about 50 M5+ flares from the testing dataset. Although this is not a very high detection rate, it shows that the model was able to identify some important flare-producing active regions despite the difficulty of the problem.

4. Overall Scientific Conclusion

This project showed that machine learning can learn useful patterns from solar magnetic field measurements and use them to predict M5+ solar flares. Throughout the project, I performed data exploration, preprocessing, feature engineering, handled class imbalance using SMOTE, trained multiple models, and evaluated them on an unseen testing dataset.

The Decision Tree model mainly learned patterns related to magnetic complexity, stored magnetic energy, and current helicity. Features such as TOTUSJH, TOTPOT, USFLUX, R_VALUE, and the engineered ratio features appeared to contribute the most to the prediction. These features are directly related to the physical properties of solar active regions, which makes the model's predictions scientifically reasonable.

The model correctly detected around 50 M5+ flares, but it also missed many flare events. One possible reason is that solar flares are highly complex physical processes. The available dataset contains only photospheric magnetic field measurements, while the actual flare develops in the Sun's corona. Important processes such as magnetic reconnection, plasma motion, and coronal magnetic field evolution cannot be directly captured by the available features.

Another limitation is that the dataset mainly contains individual observations instead of continuous time-series information. Since solar magnetic fields change continuously before a flare occurs, having information about how these parameters evolve over time could improve prediction accuracy. The severe class imbalance also made it difficult for the models to learn the minority class even after applying SMOTE.One interesting result was that the simplest model, the Decision Tree, performed better than the more advanced ensemble models in terms of Recall. This shows that a more complex algorithm does not always produce the best result. The choice of the final model should always depend on the objective of the problem rather than the complexity of the algorithm.

Overall, I do not think that missing many M5+ flares means the model is poor. I followed a proper machine learning workflow and evaluated the model using unseen data, so the final performance reflects the limitations of the available data rather than a mistake in the modeling process. Predicting powerful solar flares is still a difficult scientific problem, and even current forecasting systems face similar challenges.

In conclusion, this project helped me understand both the strengths and limitations of machine learning in space weather prediction. While the magnetic field measurements contain useful predictive information, accurately forecasting M5+ solar flares remains a challenging task. Future improvements could include time-series magnetic field data, additional solar parameters, and more advanced machine learning techniques to improve detection performance.

Observational Module: The Galactic Neighbour

Processing the M31 Andromeda Galaxy

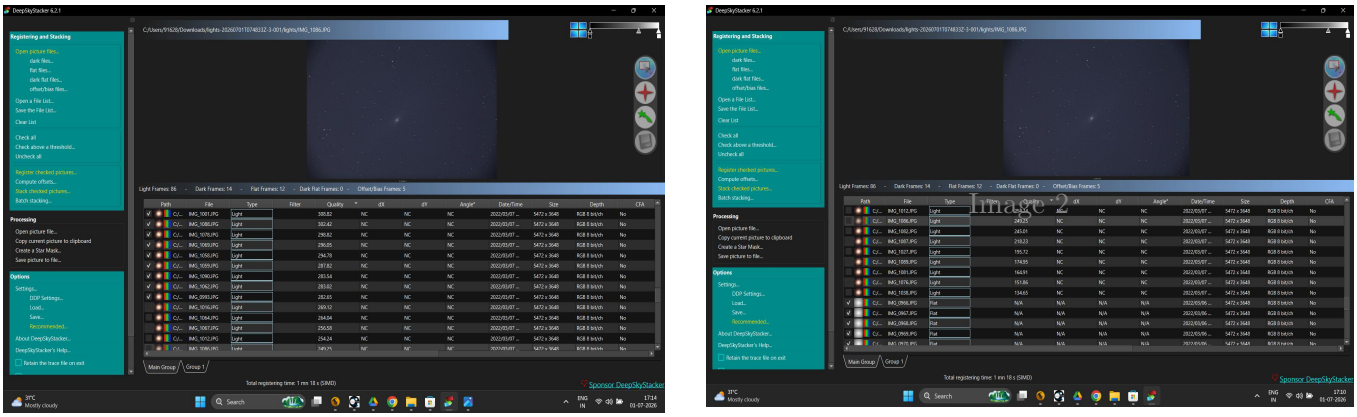
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1 Introduction

For this part of the project, I worked on processing raw telescope data of the Andromeda Galaxy (M31). The main goal was to extract the faint spiral arms and dust lanes from the raw images. Since astrophotography data always has a lot of thermal and electronic noise, the biggest challenge was to clean up the background sky without accidentally deleting the actual galaxy data.

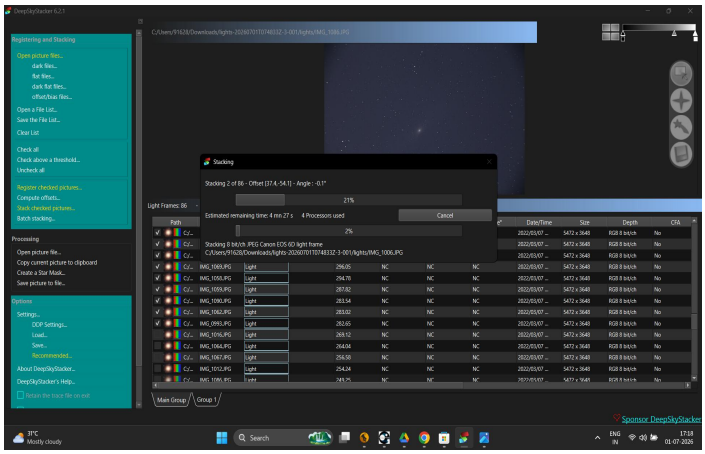
2 Task 1: Calibration & Integration (DeepSkyStacker)

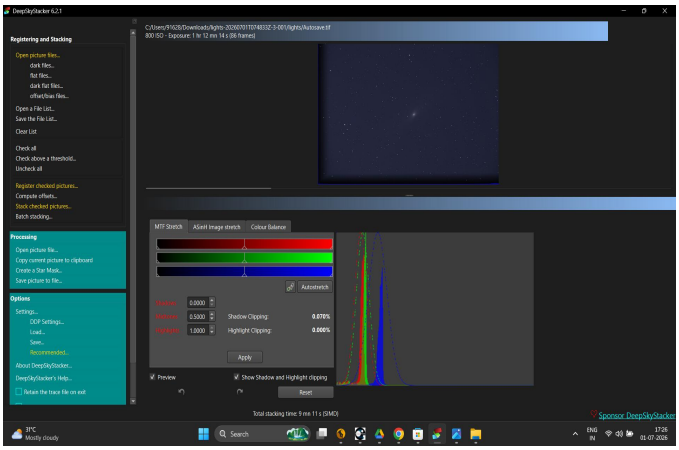
The first step was to calibrate and stack all the frames in DeepSkyStacker. The dataset I used had 86 Light frames, 14 Darks, 12 Flats, and 5 Bias frames.



Caption: Checking the frame list in DSS. I looked at the registration scores and manually unchecked the worst 10% of the light frames so that tracking errors wouldn't blur the final stacked image.

The flat frames and bias frames did a good job of removing the vignetting (dark corners) and the camera's read noise. I used the Kappa-Sigma clipping algorithm to stack everything together.

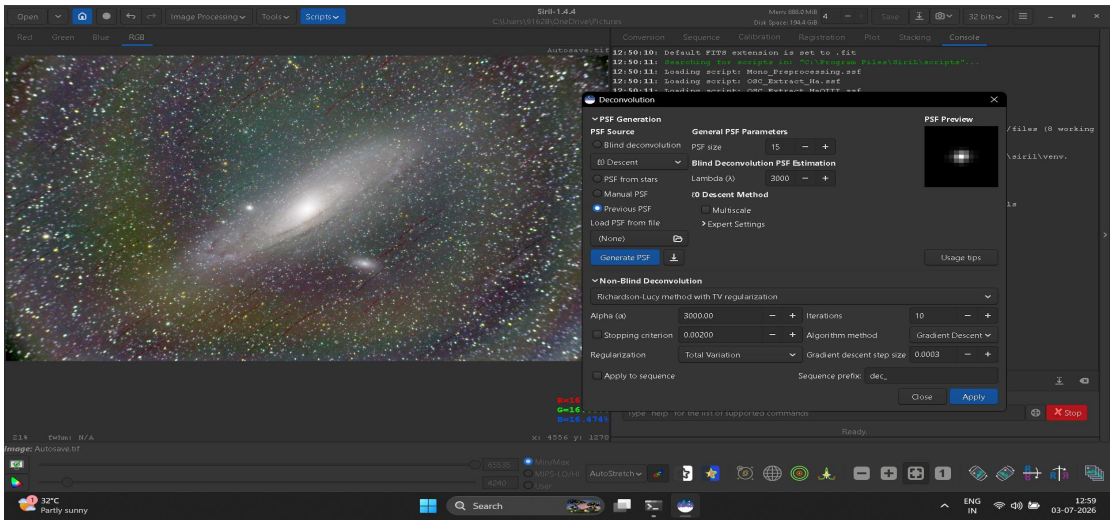




Caption: This is the final 32-bit linear image right after stacking. As expected for linear data, it is mostly dark before stretching.

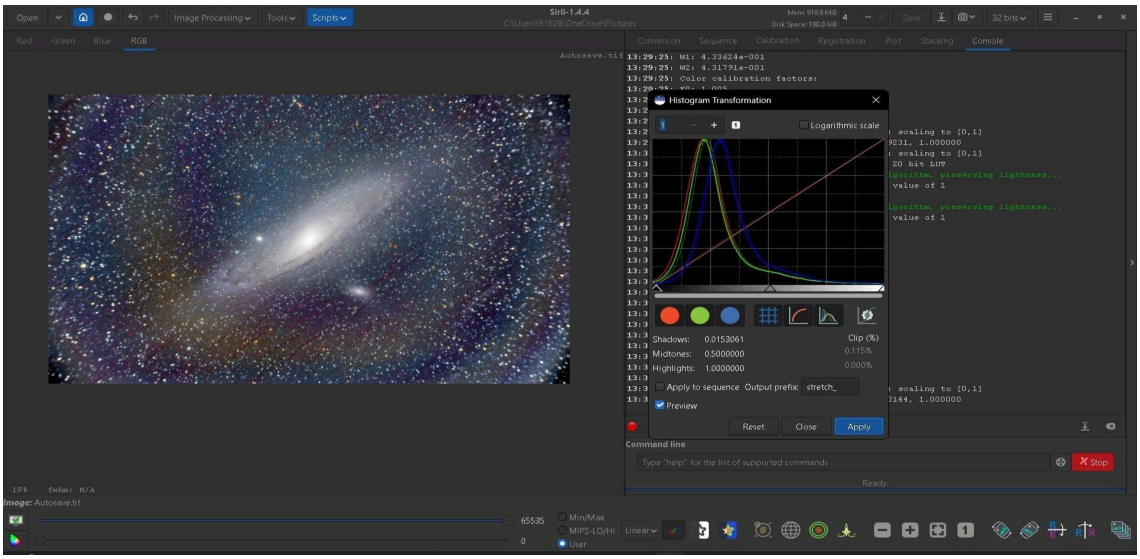
3 Task 2: Signal Extraction (Siril)

After stacking, I brought the linear TIFF file into Siril to extract the galaxy's signal. The very first thing I did was crop the edges to remove the stacking artifacts. The raw background was pretty uneven due to light pollution, so I ran a Background Extraction using the RBF (Radial Basis Function) method to flatten the sky. I also ran a Color Calibration using the background stars to make sure the galaxy's natural colors were scientifically accurate before stretching.



Caption: I applied a Richardson-Lucy Deconvolution (10 iterations) and generated a PSF to tighten up the stars so they don't look bloated.

Once the stars were fixed, I had to do a non-linear histogram stretch to make M31 visible. This part is tricky because if you push it too much, it ruins the data.



Caption: The histogram stretch window. I was very careful to keep the shadow clipping really low (at 0.115%). If we clip the shadows too much, the background turns pitch black and we permanently lose the faint data.

4 Task 3: Noise Mitigation (GIMP)

The stretching worked and the galaxy came out great, but it also exposed a lot of ugly diagonal red and purple thermal noise (called walking noise) in the background. To fix this, I used GIMP to decompose the image into the LAB color space.



Caption: Final Execution.I applied an Gaussian blur (radius 250) exclusively to the 'a' and 'b' chrominance layers. Because all the actual galaxy detail is safely locked in the unblurred L (Luminance) layer, pushing the blur this high completely averaged out the color noise across the background without ruining the sharpness of the dust lanes. This completely killed the color noise, but since I didn't touch the L (Luminance) layer, the galaxy's structure stayed perfectly sharp. After neutralizing the noise in the 'a' and 'b' layers, I used the Recompose tool in GIMP to merge the L, a, and b channels back into a standard RGB color image.

5 Artifact Analysis: The Unidirectional Black Streaks

Even though the color noise was gone, I noticed some faint, unidirectional black lines left in the background. After analyzing the data, I realized these are actually dead pixel trails.

Because the telescope was slowly drifting over the night and no mechanical dithering was used, the dead pixels on the camera sensor dragged across the frames. Even though I used Dark frames, there was likely a slight temperature difference (thermal mismatch) between when the lights and darks were shot. Because of this, DSS couldn't subtract them perfectly.

The default Kappa-Sigma clipping in DSS just wasn't strict enough to recognize these faint residual dead pixels as outliers. To tackle this in future projects, maybe i can manually lower the Kappa threshold (like setting it down to 1.5) so the software aggressively deletes these trailing dark pixels before merging the images.

6 Conclusion



Caption: The final processed image of M31. The color noise is removed, and the core and dust lanes are clearly visible.

Overall, the processing pipeline was successful in extracting the M31 signal. Using the LAB decomposition trick was really effective for neutralizing the thermal noise. Figuring out the dead pixel trails was also a great learning experience on why strict outlier rejection and matching temperatures for calibration frames are so important in astrophotography.